

Exercício 5 - Testes de Mutação

Resumo da primeira execução:

```
~/Ex5 via 🐍 v3.13.7 (env)
→ PYTHONPATH=src mutmut run ; mutmut browse
:: Generating mutants
  done in 19ms (0 files mutated, 0 ignored, 5 unmodified)
:: Listing all tests
:: Running clean tests
  done
:: Running forced fail test
  done
Running mutation testing
: 646/646 🐛 183 😬 438 🕒 0 😞 0 😞 25 🚫 0 🧑 0
202.33 mutations/second
```

Mutantes que não foram mortas na primeira execução selecionados:

Path	🐛	😬	🕒	😞	🚫	🧑	?	name
src/board.py	15	97	0	0	0	0	0	puzzle_game.xIPuzzleGameIImove_tile_from_a_posit
src/invalid_position_exception.py	4	0	0	0	0	0	0	puzzle_game.xIPuzzleGameIImove_tile_from_a_posit
src/puzzle_game.py	16	177	0	0	0	0	1	puzzle_game.xIPuzzleGameIImove_tile_from_a_posit
src/puzzle_shuffle_strategies.py	0	163	0	0	0	0	0	puzzle_game.xIPuzzleGameIImove_tile_from_a_posit
src/tile.py	0	1	0	0	0	0	0	puzzle_game.xIPuzzleGameIImove_tile_from_a_posit

Path	🐛	😬	🕒	😞	🚫	🧑	?	name
puzzle_game.xIPuzzleGameIImove_tile	20							puzzle_game.xIPuzzleGameIImove_tile mutmut 20
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d
puzzle_game.xIPuzzleGameIImove_empty_cell_to_d								puzzle_game.xIPuzzleGameIImove_empty_cell_to_d

Survived (exit_code=0): No test detected this mutant. (press t to view tests executed for this mutant)

```
--- src/puzzle_game.py
+++ src/puzzle_game.py
@@ -2,7 +2,7 @@
 tile_line = self.dic_positions_of_tiles.get(tile)[0]
 tile_column = self.dic_positions_of_tiles.get(tile)[1]
 if self.board.is_inside_the_board(tile_line, tile_column) and self.board.positions_are_adjacent(tile_line, tile_
- self.board.put_tile(tile, self.line_of_empty_position, self.column_of_empty_position)
+ self.board.put_tile(None, self.line_of_empty_position, self.column_of_empty_position)
 self.dic_positions_of_tiles.update({tile: (self.line_of_empty_position, self.column_of_empty_position)})
 self.line_of_empty_position = tile_line
 self.column_of_empty_position = tile_column
```

Path	🟡	🟠	🔴	🟡	🔴	🟡	🔴	?	name
src/board.py	15	97	0	0	0	0	0	0	puzzle_game.xl PuzzleGame _print_board_of_puzzl
src/invalid_position_exception.py	4	0	0	0	0	0	0	0	puzzle_game.xl PuzzleGame _print_board_of_puzzl
src/puzzle_game.py	14	177	0	0	0	0	0	1	puzzle_game.xl PuzzleGame _print_board_of_puzzl
src/puzzle_shuffle_strategies.py	0	163	0	0	0	0	0	0	puzzle_game.xl PuzzleGame _print_board_of_puzzl
src/tile.py	0	1	0	0	0	0	0	0	puzzle_game.xl PuzzleGame _print_board_of_puzzl

```
Survived (exit code=0): No test detected this mutant. (press t to view tests executed for this mutant)
```

Path	🙄	👴	🚫	😞	🔒	👨	🔴	?	name
src/board.py	15	97	0	0	0	0	0	0	puzzle_game.xlPuzzleGame1_print_board_of_puzzle
src/invalid_position_exception.py	4	0	0	0	0	0	0	0	puzzle_game.xlPuzzleGame1_print_board_of_puzzle
src/puzzle_game.py	14	177	0	0	0	0	0	1	puzzle_game.xlPuzzleGame1_print_board_of_puzzle
src/puzzle_shuffle_strategies.py	0	163	0	0	0	0	0	0	puzzle_game.xlPuzzleGame1_print_board_of_puzzle
src/tile.py	0	1	0	0	0	0	0	0	puzzle_game.xlPuzzleGame1_print_board_of_puzzle

Survived (exit code=0): No test detected this mutant. (press t to view tests executed for this mutant)

Resultado com novos testes:

```
~/Ex5 via 🐍 v3.13.7 (env) took 6s
→ PYTHONPATH=src mutmut run
:: Generating mutants
    done in 20ms (0 files mutated, 0 ignored, 5 unmodified)
:: Listing all tests
:: Running clean tests
    done
:: Running forced fail test
    done
Running mutation testing
:: 646/646 🧑‍🚒 183 🕒 438 🕒 0 🧐 0 🧐 25 🚫 0 🧑‍🚒 0
192.16 mutations/second
```

Outros Mutantes que continuaram vivos:

Path		😊	🤖	🔴	🟡	⚠️	🟢	❓		statu
src/board.py	13	97	0	0	0	0	0	0	move_empty_tile mutmut_5	
src/invalid_position_exception.py	4	0	0	0	0	0	0	0	move_empty_tile mutmut_6	
src/puzzle_game.py	8	177	0	0	0	0	0	0	move_empty_tile mutmut_7	
src/puzzle_shuffle_strategies.py	0	163	0	0	0	0	0	0	move_empty_tile mutmut_8	
src/tile.py	0	1	0	0	0	0	0	0	move_empty_tile mutmut_9	
									move_empty_tile mutmut_10	
									move_empty_tile mutmut_11	
									move_empty_tile mutmut_12	
									move_empty_tile mutmut_13	
									move_empty_tile mutmut_14	
									move_empty_tile mutmut_15	
									move_empty_tile mutmut_16	
									_print_board_of_puzzle_game mutmut_1	
									_print_board_of_puzzle_game mutmut_2	
									_print_board_of_puzzle_game mutmut_3	
									_print_board_of_puzzle_game mutmut_4	
									_print_board_of_puzzle_game mutmut_5	
									_print_board_of_puzzle_game mutmut_6	
									_print_board_of_puzzle_game mutmut_7	
									_print_board_of_puzzle_game mutmut_8	
									_print_board_of_puzzle_game mutmut_9	
									_print_board_of_puzzle_game mutmut_10	
									_print_board_of_puzzle_game mutmut_11	
									_print_board_of_puzzle_game mutmut_12	
									_print_board_of_puzzle_game mutmut_13	
									_print_board_of_puzzle_game mutmut_14	
									_print_board_of_puzzle_game mutmut_15	
									_print_board_of_puzzle_game mutmut_16	
									_print_board_of_puzzle_game mutmut_17	
									_print_board_of_puzzle_game mutmut_18	
									_print_board_of_puzzle_game mutmut_19	
									_print_board_of_puzzle_game mutmut_20	
									_print_board_of_puzzle_game mutmut_21	
									end of the game mutmut_1	
									get_tile mutmut_10	😊